



STAGEMARK 2027

SiLeMi/o · By Mamat · -----[]--

User manual

Metric stage marking, plans, projector calibration and safe output control.

SHOW RULE - The output always opens black. Verify the selected display, then use **SHOW**. **BLACKOUT** remains available and takes priority at all times.

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Windows 11 and macOS · local and offline operation

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This manual follows the recommended path: metric stage → objects → plans → file → calibration → projection → blackout. Persisted coordinates always use integer millimetres.

1. Output safety principles

StageMark prepares and projects vector marks on a stage. It does not control the projector hardware: it uses a secondary display connected to the video system on Windows or macOS, or a Spout 2 source received by mapping software on the same Windows PC.

- No projection window exists at startup.
- Opening the output first creates a fully black, borderless surface with no pointer.
- Any display removal or metric change immediately forces blackout.
- The reference background imported into the editor is never sent to the projector.
- BLACKOUT also disarms the local controller's Instant point.
- Spout output must be started explicitly on the PC. It opens black and sends another black frame before stopping.

Before a show: disable notifications, sleep, screen savers and Windows updates; secure the cable path; test BLACKOUT from the PC, phone and Stream Deck.

2. Quick start

1. In Stage, enter the real width and depth, then choose the outline and grid.
2. In Drawing, create marks and use numeric fields for exact positions and dimensions.
3. Create or choose the working Plan, then save a standalone StageMark .stagemark project, open a local StageFlow .stageflow project, or explicitly join a StageFlow Network LIVE session.
4. In Projection, choose and open the display, or start Spout. Every output stays black.
5. In Installation, choose the projector, then show the test pattern and align its four corners with the real target.
6. Return to the plan and select SHOW. Use BLACKOUT whenever no image should be visible.

SHOW is disabled until the output is open. BLACKOUT is always available from every workspace.

3. Metric stage and navigation

The origin (0, 0) is at centre stage on the downstage edge. Negative X is stage right/Jardin, positive X is stage left/Cour, and positive Y points upstage. Metres are only a display conversion: project data remains in integer millimetres.

- Project metadata (name, venue, author, event date and notes) is edited in Stage and feeds team outputs.
- The display-unit selector only changes on-screen readings; stored coordinates remain integer millimetres.

Action	Gesture
Zoom	Mouse wheel
Pan the view	Ctrl + drag the background
Alternative pan	Alt + drag or middle mouse button
Precise move	Arrows: 10 mm · Shift + arrows: 100 mm
Add to selection	Ctrl + click · Shift + marquee

Scaled reference plan

- Import a PNG, JPEG, WebP or SVG: the image is embedded in the project and remains strictly editor-only.
- Unlock it and choose Edit on stage. Drag the image to move it; drag a side for a free dimension or a corner to preserve proportions.
- Choose Scale by measure, click two points whose distance is known, enter the real distance in metres and apply. StageMark scales proportionally around the first point, using integer millimetres.

4. Objects, free shapes and plans

Objects belong to the active plan. The permanent Site / obstacles layer is shared by all plans and excluded from projection by default. Edit groups move objects together; display snapshots only store their visibility.

- Create, rename, lock and hide layers, then reassign objects without changing their geometry.
- Distribute a selection along X or Y for regular spacing; existing alignment tools remain available.
- Dimension mode measures a stage distance and can keep a non-projected technical annotation.

Free shapes and Pen tool

- Click: create a corner and a straight segment.
- Click-drag: create a smooth point and Bézier handles.
- Straight and curved gestures can be mixed within one path.
- Enter closes a shape; Shift + Enter finishes an open line; Backspace removes the last point.
- After creation, Edit path can add, remove or convert a point between Corner / straight and Smooth curve.

Plans and visibility

- Create, duplicate, rename, reorder, hide or delete a plan in the left column.
- Create a snapshot from a selection, then Load, Close or Solo it.
- Obstacles + plan or Obstacles only must be selected explicitly during preparation.

Do not use the reference background as a stage mark. It is deliberately removed from every projection payload.

5. Standalone, local StageFlow and Network LIVE projects

One show, three explicit access paths

StageMark	STANDALONE STAGEMARK PROJECT Name.stagemark	No StageFlow required
Suite SiLeMIO	LOCAL STAGEFLOW PROJECT Name.stageflow	Folder accessible on this computer
StageFlow free and optional	STAGEFLOW NETWORK LIVE SESSION	LAN discovery or private IPv4 + 6-digit code

StageMark distinguishes three access paths. A .stagemark file is the complete standalone StageMark project. A .stageflow folder accessible on this computer is a local StageFlow project that StageMark can create, open and save without running StageFlow. A StageFlow Network LIVE session is broadcast by StageFlow on the local network, discovered in StageMark, then joined voluntarily with its 6-digit code. StageFlow is free and optional.

Network LIVE does not share the host computer's path and uses neither cloud storage nor a fake local file. StageMark validates the remote project and its assets in memory before enabling following or commands. After the initial snapshot, every LIVE revision waits for an explicit operator action, even while the editor is clean; it can never show an image, remove BLACKOUT, change the projection display or recall a cue.

- In a standalone .stagemark project, Save writes the complete V11 model to the same file. Save As creates a new .stagemark file and leaves the source untouched.
- In a local .stageflow project, Save writes only stagemark/stagemark.json. Locks, the base hash, manifest merging and atomic replacement protect every foreign domain, including Patch, CAD, StageDesk, StageMon and Dante Config Editor.
- Open Connection center (top-right icon or Ctrl+L), choose StageFlow LIVE session, Refresh sessions, the project and host computer, then enter the 6-digit code. Leading zeroes are preserved; ASCII spaces and hyphens are accepted, but letters and punctuation are never silently repaired. If multicast discovery fails, enter the StageFlow computer's private IPv4 address as a fallback. Pairing is never automatic.
- The network project stays in memory only: no cache, autosave or recovery copy is written to disk. The manual address, code and token are not persisted, and StageMark never reconnects by itself.
- Saving in Network LIVE publishes only the StageMark domain with its base hash. A stale base creates a visible conflict and preserves local edits instead of overwriting the remote project.
- The local-project watcher and network following validate projectId, revision and hash. A partial file or invalid response never replaces the last valid state; every later revision waits for an explicit operator action.
- Network following and commands start only after the exact project is installed. A different session or project identity is rejected. Explicit show-control commands remain separate from project updates.
- Detaching from Network LIVE, losing the session or failing during activation keeps the drawing as a modified, unsaved standalone StageMark copy.
- The bundled AutoCAD connector works with a local StageFlow project and can directly join a remote LIVE session; this path was validated in real AutoCAD with StageFlow commit dd79337.
- Alerts are ON at every join: an orange operator banner shows previous → current value, label/group, author and time. Acknowledge or Acknowledge all on this workstation targets alerts visible at the click; later changes remain pending. View list opens the details.

- Local reception OFF has no backlog when enabled again. StageFlow's notification pause is shown separately: LIVE, synchronisation and safety continue. Neither setting remotely acknowledges alerts belonging to another workstation.
- Save and Save As always preserve the active project type. Copy commands between .stagemark and .stageflow are explicit and non-destructive, and leave the source untouched.
- Import legacy... opens an old .json file without changing it; its first save creates a .stagemark file. An existing .stagemark file opens directly as a standalone project.
- CadOverlay remains separate from the reference image and its projection is off by default. An update accepted in the editor is never sent to an output without an operator action.
- Local autosave allows recovery on the next launch after a local-project interruption; it is disabled for a Network LIVE session.
- Recent projects reopens a .stagemark file, a .stageflow folder or an imported legacy JSON, removes a missing entry or clears only the history. No project is deleted.
- Show projects and personal calibrations must never be committed to the source repository.
- Ctrl+S saves; Ctrl+Shift+S chooses a new local destination of the same type; Ctrl+O offers a standalone project, a local StageFlow project or a StageFlow Network LIVE session.

PDF, PNG and printing

- Choose one or more plans, then inspect the preview before producing any output.
- Enable grid, site/obstacles, reference background, legend, scale and dimensions independently for the team document.
- PDF and printing can create multiple A4 pages, one plan per page. The suggested orientation can always be overridden.
- PNG exports one plan. The filename includes the project, plan and timestamp.

6. Installation, calibration, projection and cueing

Installation is deliberately separate from show operation. Its Projector, Placement and Calibration tabs bring together the equipment bank, lenses, calculations and geometry settings.

1. Connect the projector and select an extended display in Windows or macOS settings, at native resolution.
2. In Projection, explicitly select the projector display.
3. Select Open output. The chosen display turns black.
4. In Installation, verify the projector and lens, then select Show test pattern in Calibration.
5. Move the four numbered corners. Arrows: 1 px; Shift + arrows: 10 px.
6. When the pattern matches the real target, use SHOW for the plan and BLACKOUT to cut it.

Calibration profiles

- Create, rename and duplicate a profile for each real installation; a duplicate must be validated again on site.
- Associate the profile with the Windows display, projector and lens in use. A technical change automatically invalidates the previous validation.
- Select Validated on site only after physically checking the pattern and blackout. The final profile cannot be deleted.

Calibration corrects reasonable perspective, but cannot replace suitable optics. Strong side or oblique projection reduces focus, brightness and usable resolution.

Obstacle modes

Mode	Projected content
Off	Active plan objects only
Obstacles + plan	Active plan and permanent Site objects
Obstacles only	Permanent Site objects, with the plan emptied

Local Spout output (Windows only)

For MadMapper, OBS or another Spout receiver on the same PC, choose the sender name, resolution and frame rate, then select Start in BLACKOUT. SHOW, BLACKOUT, the test pattern, plans and Instant point control the direct display, Spout, or both. The first version uses a BGRA CPU copy; check motion and blackout latency at the production resolution.

Cueing and preflight

- Capture cues from the current state, rename and reorder them, then recall them during operation.
- Preflight checks the essentials before SHOW: output, display, calibration, plan and safety state.
- Recalling a cue never bypasses priority blackout. Prepare and rehearse the complete cue list before the audience enters.

7. Local phone controller

In Connection center → StageMark remote, choose the network and explicitly start the controller. Scan its QR code on a device connected to the same private network. The session uses an ephemeral token and permits only one active operator.

During a StageFlow Network LIVE session, the optional StageFlow QR portal can open this exact same remote. StageMark must have started the local controller first; the handoff is temporary, does not disclose the primary token and never triggers output. StageMark's direct QR remains available without StageFlow. When StageFlow and StageMark use different IP addresses, the portal opens the remote full-screen; browser Back returns to StageFlow. Opening a

connection screen never starts a server or output.

- The phone can open direct display output, close all active outputs, show the plan, engage blackout and show the test pattern.
- The Calibration section in Installation adjusts all four corners from the real target surface.
- Instant point projects a temporary mark without changing the file. Blackout removes it.
- The phone can also create and edit essential marks and plans.
- The phone Cueing tab can choose a plan, recall a cue and inspect preflight while standing on stage.

The controller uses no cloud or account. It is powerful: share the session QR only with the authorised operator and stop the session after use.

8. Projector placement calculator

The placement calculator sits in Installation, with the projector bank and immediately before calibration. It handles a projector centred above the stage, at front of house, or outside either stage side.

Useful width = $\max(\text{stage width, depth} \times \text{image aspect ratio}) \times \text{margin}$.

Lens-to-floor height = useful width $\times$ throw ratio.

Input	Source
Resolution	Projector native resolution, for example 1920 $\times$ 1200
Minimum / maximum throw	Manufacturer data for the lens actually fitted
Margin	Usually 5 to 15% around the stage

The computed height is lens-to-floor, not rigging-point height. Physically confirm zoom, focus, brightness, shadows, structural load and safe suspension.

For an oblique position, StageMark also estimates centre throw distance, effective throw ratio, angle and perspective spread. Homography cannot restore lost pixels, brightness or depth of field; the final decision belongs to the video and rigging professionals.

The bundled catalog contains ten professional projectors and 100 lenses. Search, filters, sorting and local favourites simplify selection. Applying a record copies only resolution and throw range into the calculator. Catalog V2 verifies its manifest and SHA-256 before import. A photo is shown only when its licence explicitly permits redistribution; no unlicensed manufacturer image is bundled.

9. Stream Deck and shortcuts

Key	Action
F13	Priority BLACKOUT
F14	SHOW the plan
F15 / F16	Previous / next plan
F17 / F18	Open / close output
F19	Show test pattern
F20	Arm Instant point
F21 / F22	Previous / next cue
Ctrl+Alt+1 to 8	Recall cues 1 to 8 directly

In Stream Deck, assign the Hotkey action to each key. No StageMark plugin, account or cloud service is required.

OSC and MIDI controls

Open Tools → OSC · MIDI · Stream Deck to configure OSC input and feedback, select MIDI ports, learn a message and open the Stream Deck kit. StageMark stays standalone; in LIVE, StageFlow relays the same identifiers.

ID	OSC address
stagemark.output.show	/silemio/stagemark/output/show
stagemark.output.blackout	/silemio/stagemark/output/blackout
stagemark.cue.select	/silemio/stagemark/cue/select
stagemark.cue.next	/silemio/stagemark/cue/next
stagemark.cue.previous	/silemio/stagemark/cue/previous

10. Manual, license and updates

Open this PDF from Help. Trial and license displays the local status and accepts a signed license code.

Trial and permanent license

- The first 30 days remain fully usable with no reminder.
- After 30 days, every feature remains available. Only a dismissible reminder is shown.
- The permanent license is a one-time €49 including tax purchase. It removes the reminder and remains valid on Windows, macOS and compatible updates.
- Purchase opens explicitly in the browser. Existing V1 licenses remain portable and verified offline. A V2 license needs a first online activation for each installation, then works offline with its signed receipt.
- Commercial V2 licenses include three installations by default. Trial and license shows the count and lets you deactivate this computer. No StageMark account, recurring online check or telemetry is required.

Licensing has no access to the stage, files, calibration, projection, local controller or blackout. A licensing error never blocks operation.

Safe updates

At startup, StageMark may check the latest Release in the public GitHub distribution repository, without an account or embedded token. Download requires your approval, the installer must match its SHA-256 file, and launching it requires a second explicit action. Launching forces blackout, closes output and stops the local controller.

An update is never installed silently and must never be launched during a show.

Quick troubleshooting

Symptom	Check
Output stays black	Is output open? Is SHOW active? Is the correct display selected?
Wrong geometry	Show the pattern again, restore corner order and avoid crossing the quadrilateral.
Phone unavailable	Same private network, session started, current QR, Windows Firewall allowed.
Update unavailable	Check Internet, then open the public Release manually if necessary.
Installer rejected	Download again: StageMark never accepts a missing or mismatched SHA-256.
License code rejected	Copy the complete code again. Validation is local and the app remains usable.

License and help

StageMark remains fully usable without a license. The one-time €49 including tax license removes the reminder after the 30-day trial. It is personal, has no subscription, and remains valid through compatible updates. V1 licenses remain portable; V2 licenses use installation activations, which can be released in Trial and license.

Buy a license

<https://stagemark-license.mamat79-dce.workers.dev/buy>

Stripe processes payment in the browser. StageMark does not receive, store or transmit bank details. The license code is then verified offline.

SiLeMi/o suite

StageFlow and its AutoCAD connector are free and optional, with no SiLeMIO activation. StageMark works independently with .stagemark, opens .stageflow locally without StageFlow, and voluntarily joins LAN LIVE with a 6-digit code, without a shared path or cloud. The local console (presence and pipes) is Windows-only, separate from network LIVE and the remote controller. AutoCAD keeps its separate Autodesk license.

StageFlow · StageDesk · StageMark · StageMon · Dante Config Editor · AutoCAD

Downloads and issues

<https://github.com/Mamat79/StageMark>

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Thank you for using StageMark. Always run a complete rehearsal with the projector, Windows display, remote controller and safety controls before the audience enters.